

1. Administrative & Study Identification

Full Study Title: A Phase I, Multicentre Study to Assess the Safety, Tolerability, and Pharmacokinetics of Ascending Doses of AZD1390 in Combination with Radiation Therapy in Patients with Glioblastoma Multiforme and Brain Metastases from Solid Tumors
Short Title/Acronym: A Study to Assess the Safety and Tolerability of AZD1390 Given With Radiation Therapy in Patients With Brain Cancer
Protocol Number: D6940C00002 **NCT Number:** NCT03423628
Sponsor Name: AstraZeneca **Investigational Product:** AZD1390 (ATM Kinase Inhibitor)
Phase of Development: Phase I
Medical Monitor: AstraZeneca Clinical Operations Lead / Independent Data Monitoring Committee (DMC)

2. Study Synopsis & Rationale

2.1 Study Objective
Primary Objective: To assess the safety and tolerability of ascending doses of AZD1390 given in combination with radiation therapy (RT) in patients with brain cancer and to determine the maximum tolerated dose (MTD).
Secondary Objectives: To evaluate preliminary efficacy including Overall Survival (OS) and to characterize the pharmacokinetics (PK) of AZD1390 (Cmax, AUC, half-life).

2.2 Background & Rationale
Glioblastoma multiforme (GBM) is a highly aggressive and lethal brain cancer with poor long-term survival outcomes. A primary reason for treatment failure is the cancer cells' ability to repair DNA damage caused by standard-of-care radiation therapy. AZD1390 is a novel, blood-brain barrier-penetrant ataxia telangiectasia mutated (ATM) kinase inhibitor. By inhibiting ATM—an enzyme critical for DNA damage repair—AZD1390 acts as a radiosensitizer to potentiate the efficacy of radiation therapy in brain tumors without exacerbating neurotoxicity in surrounding healthy brain tissue.

3. Study Design & Methodology

3.1 Design Framework
Study Type: Interventional
Control Type: Single Group Assignment (Dose Escalation/Dose Expansion)
Blinding: Open-label
Randomization: N/A

3.2 Planned Enrollment & Duration
Target N\$: 180 (Estimated Enrollment)
Number of Sites: Multicentre (Sites across the United States, United Kingdom, and Japan)
Estimated Study Start: 02-Apr-2018
Estimated Primary Completion: 16-Sep-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Karnofsky Performance Score ≥ 60 . 2. Histologically proven diagnosis of solid tumor malignancy and Magnetic Resonance (MR) imaging documenting brain lesions. 3. For Arm A: A radiological diagnosis of recurrent/relapsed or progressive disease according to RANO criteria, with completion of first-line radiation at least 6 months prior. 4. Willingness to receive anti-epileptic prophylaxis for the duration of study drug administration.

4.2 Exclusion Criteria 1. Administration of chemotherapy or any investigational drug in the 28 days or carmustine (CCNU) or lomustine (BCNU) in the 6 weeks prior to receiving the first dose of treatment in Arms A and C. 2. Patient has received any previous brain RT to the area that is to be irradiated. 3. History of severe brain injury or stroke. 4. Patient not eligible for sequential MRI evaluations.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: AZD1390 is administered in ascending once-daily doses across 3 treatment arms. * **Arm A (Recurrent GBM):** 35 Gy over 2 weeks with intensity-modulated radiation therapy (IMRT). AZD1390 administered in 3 cycles: Cycle 0 (1 dose prior to RT), Cycle 1 (during RT), Cycle 2 (2 weeks adjuvant). * **Arm B (Brain Metastases):** 30 Gy WBRT/PBRT over 2 weeks (Arm closed to recruitment). * **Arm C (Primary GBM):** 60 Gy over 6 weeks with IMRT, plus AZD1390. **Route of Administration:** Oral (AZD1390) and External Beam (Radiation Therapy). **Duration of Treatment:** Up to 8 weeks of treatment (depending on the arm, e.g., 6 weeks concurrent with RT + 2 weeks adjuvant in Arm C).

5.2 Concomitant & Prohibited Therapy Permitted: Anti-epileptic prophylaxis (mandatory). **Prohibited:** Chemotherapy, checkpoint inhibitors (within 28 days prior), concurrent therapy with other seizurogenic medications.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Histological Confirmation, Baseline MRI, Karnofsky Score
Baseline (Cycle 0)	Day 0	Single dose of AZD1390 prior to RT, Pharmacokinetic (PK) Analysis
Interim (Cycle 1)	Weeks 1 - 6	Daily AZD1390 dosing concurrent with Radiation Therapy, Safety Labs
Adjuvant (Cycle 2)	Weeks +1 to +2	2 weeks adjuvant AZD1390 treatment post-RT (Arms A and C)
End of Study	Long-term	Safety follow-up, OS/Efficacy assessment post-treatment

(Note: Timelines vary slightly depending on group assignment to Arm A, B, or C).

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Safety and tolerability profile of AZD1390, determined by the incidence and severity of Adverse Events (AEs) and Dose-Limiting Toxicities (DLTs) to establish the Maximum Tolerated Dose (MTD). **Measurement Method:** Clinical evaluation, laboratory monitoring, and grading using NCI CTCAE v4.03.

7.2 Secondary & Exploratory Endpoints * **Preliminary Efficacy:** Overall Survival (OS). * **Pharmacokinetics:** Plasma concentrations of AZD1390 including C_{max} , AUC , and elimination half-life.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous clinical and laboratory monitoring with a specific focus on identified risks such as radiation skin injury, fatigue, headache, nausea, and skeletal muscle toxicity (e.g., creatine kinase elevation). **Serious Adverse Events (SAEs):** Reported standardly to identify DLTs; DLTs in early cohorts define the dose escalation rules for subsequent cohorts. **Data Monitoring Committee (DMC):** An independent group of scientists monitoring the safety and scientific integrity of the trial, with the authority to recommend trial cessation if deemed unsafe.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all enrolled patients receiving at least one dose of AZD1390). **Sample Size Justification:** $N=180$ is scaled appropriately for a Phase 1 dose-escalation and expansion study to determine MTD and gather robust PK and preliminary OS data across multiple clinical cohorts (Arms A, B, C). **Handling of Missing Data:** Standard Phase 1 imputation techniques applied for PK/PD parameters and safety readouts.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Scheduled onsite and remote clinical monitoring for protocol adherence, toxicity grading, and PK/PD sample integrity. **Audit Trail:** Maintained via robust electronic Case Report Form (eCRF) platforms tracking dose escalations and radiological (RANO criteria) evaluations.

11. Study Identifiers & Governance

Primary ID: D6940C00002 **Registry Number:** NCT03423628 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** A Study to Assess the Safety and Tolerability of AZD1390 Given With Radiation Therapy in Patients With Brain Cancer **Official Regulatory Title:** A Phase I, Multicentre Study to Assess the Safety, Tolerability, and Pharmacokinetics of Ascending Doses of AZD1390 in Combination with Radiation Therapy in Patients with Glioblastoma Multiforme and Brain Metastases from Solid Tumors

12. Clinical Framework (Oncology/Brain Cancer Specific)

Primary Condition: Glioblastoma Multiforme (GBM) and Brain Metastases **Diagnosis Threshold:** Histologically proven diagnosis of solid tumor malignancy with MRI-documented brain lesions **Medical Methodology:** Intensity-Modulated Radiation Therapy (IMRT) / Whole Brain Radiation Therapy (WBRT) **Optimization Target:** Identification of the Maximum Tolerated Dose (MTD) of the radiosensitizer (AZD1390)

13. Study Procedures & Methodology

Management Pathway: Concurrent Phase 1 Dose-Escalation of oral ATM-inhibitor with Standard of Care Radiation Therapy **Education Protocol (HCP):** Clinical investigator training on dose-escalation limits, DLT reporting, and PK sample collection **Education Protocol (Patient):** Strict guidelines on fasting/fed states for the PK food-effect cohort and mandatory anti-epileptic medication compliance **Quality Audit Mechanism:** Independent Data Monitoring Committee (DMC) safety reviews

14. Observation & Follow-Up Schedule

Total Duration: Approx. 8 weeks of active intervention plus long-term survival follow-up **Onsite Visit Cadence:** Pre-treatment, continuous weekly monitoring during active RT phases, and immediate post-RT evaluations **Remote Monitoring Frequency:** Routine survival sweeps post-treatment completion **Checklist Review Interval:** Continuous AE/SAE monitoring and tumor progression scans per RANO criteria

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	____	[DD-MMM-YYYY]				

Clinical Operations Lead | [Name] | ____ | [DD-MMM-YYYY] | | Lead
Statistician | [Name] | _____ | [DD-MMM-YYYY] |